

Indian Oil Corporation Ltd



GUWAHATI REFINERY ,ASSAM



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Guwahati Refinery L&D Centre

SUMMER INTERNSHIP REPORT

(15th DEC 2025 to 15th JAN 2026)

ELECTRICAL DEPARTMENT

Submitted By :

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B.Tech 3rd Year (5th sem)

Electrical Engineering

NIT Agartala

IOCL GUWAHATI REFINERY, NOONMATI, ASSAM, INDIA

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PREFACE

Theoretical knowledge lays the foundation, but it is through practical application that true understanding is achieved. This is especially true in the field of Electrical Engineering, where hands-on experience is essential to bridge the gap between classroom learning and real-world challenges.

As a B.Tech student in Electrical Engineering at National Institute of Technology Agartala, I had the opportunity to undergo a one-month internship at I.O.C.L., Noonmati (Guwahati Refinery). This internship was conducted from 15 Dec 2025 to 6 Jan 2026. During this period, I was exposed to various practical aspects of the refinery, particularly in the Electrical Department and P&U (Power and Utilities) Department. This internship allowed me to observe and understand the functioning of several critical systems and technologies in a live industrial environment.

This experience not only enhanced my technical knowledge but also boosted my confidence and understanding of how theoretical concepts are implemented in actual industrial scenarios. The ever-evolving nature of technology makes such exposure vital for young engineers and future professionals.

I express my sincere gratitude to Mr. Soumen Chakraborty (Maintenance Manager, Electrical Engineer, IOCL Guwahati Refinery) as my instructor, for his valuable guidance and constant support throughout this internship. His mentorship played a significant role in making this training a productive and enriching experience.

I consider this internship a significant milestone in my academic and professional journey and am thankful for the learning it provided.

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ACKNOWLEDGEMENT

I consider myself fortunate to have had the opportunity to undergo a one-month internship at Indian Oil Corporation Limited (IOCL), Guwahati Refinery, from 15 Dec 2025 to 6 Jan 2026. This internship provided me with invaluable practical exposure and hands-on experience in the Electrical Department and P&U (Power and Utilities) Department, contributing significantly to my academic and professional growth.

I would like to express my sincere gratitude to Mr. Surendra Mohan (S.M.) Deka, Manager (Learning & Development, IOCL Guwahati Refinery, Noonmati), Mr. Aditya Sharma, Manager (Fire & Safety), and Ms. Shweta Mishra, STSM, for their valuable support and guidance during the course of my training.

My instructor throughout this internship was ,Mr. Soumen Chakraborty (Maintenance Manager, Electrical Engineer, IOCL Guwahati Refinery) whose mentorship and technical supervision played a vital role in making this internship a meaningful learning experience. His constant support and encouragement helped me understand the practical applications of various systems and processes.

I am also thankful to all the officers, engineers, and staff members who extended their cooperation and shared their knowledge with patience. Their contributions helped me deepen my technical understanding and gain insights into real-world industrial operations.

This internship has been a major milestone in my journey as a B.Tech Electrical Engineering student at National Institute of Technology Agartala. I will strive to apply the knowledge and skills gained during this period in my future endeavors and continue working towards my professional goals.

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ABBREVIATIONS

Abbreviation	Full meaning	Abbreviation	Full Meaning
IOCL	Indian Oil Corporation Limited	RPM	Rotation per minute
TPS	Thermal power station	Pf	Power factor
MW	Mega-watt	DPT	Differential pressure transmitter
DM	De-mineralization	Pa	Pascal(unit for pressure)
TG	Turbo-generator	Hg	Mercury
STG	Steam turbo generator	MLG	Magnetic level gauge
MP	Medium pressure	I/P	Current to pressure
LP	Low pressure	OC	Open circuit
TPH	Tons per hour	SC	Short circuit
IJT	Isgec John Thompson	RTD	Resistance temperature detector
CDU	Crude distillation unit	HV/HT	High Voltage/High tension
DCU	Delayed coking unit	LV/LT	Low voltage/Low tension
HGU	Hydrogen generation unit	CT/VT/PT	Current/Voltage/Potential Transformer
HDT	Hydrotreater unit	IDMTL	Inverse definite minimum time lag

Overview of Indian Oil Corporation Limited (IOCL)

Indian Oil Corporation Limited (IOCL) is India's largest commercial enterprise and a Maharatna Public Sector Undertaking under the Ministry of Petroleum and Natural Gas, Government of India. Established in 1959 and headquartered in New Delhi, IOCL is a fully integrated energy company operating across the entire hydrocarbon value chain, including refining, pipeline transportation, petroleum product marketing, exploration and production of crude oil, natural gas marketing, petrochemicals, and renewable energy. With over 33,000 employees and a strong pan-India presence, the company is also expanding globally, with offices in the UAE, Sri Lanka, Bangladesh, and Mauritius. IOCL owns and operates 11 refineries across India, including in Panipat, Mathura, Gujarat, Barauni, Haldia, Digboi, Guwahati, Bongaigaon, and Paradip, along with its subsidiary Chennai Petroleum Corporation Ltd. As of 2025, the company has a total refining capacity of 80.55 million metric tons per annum (MMTPA), making it the largest refiner in India. It also operates the country's largest pipeline network of over 17,000 km for the transportation of crude oil, refined products, and natural gas. Its vast marketing infrastructure includes over 59,000 retail fuel outlets, 30,000+ LPG distributors, and more than 120 aviation fuel stations. Financially, IOCL reported a revenue of ₹8.2 lakh crore and a net profit of ₹34,781 crore for the fiscal year 2023–24, and it ranks 94th on the Fortune Global 500 list (2023). The company's R&D Centre in Faridabad is a global leader in fuel, lubricant, and petrochemical innovation, holding several international patents. IOCL is also at the forefront of sustainability, with major investments in green hydrogen, ethanol, biodiesel, solar and wind energy, EV charging, and compressed biogas, while aiming for net-zero operational emissions (Scope 1 and 2) by 2046. Looking ahead, IOCL envisions transforming from a traditional petroleum company into a diversified, digitally enabled, and environmentally responsible energy corporation with plans to expand refining capacity to over 100 MMTPA by 2030 and become a global leader in green energy solutions. With its continuous innovation, commitment to nation-building, and drive for sustainability, IOCL remains a cornerstone of India's energy security and industrial growth.

IOCL GUWAHATI REFINERY, NOONMATI, ASSAM, INDIA

Overview of IOCL Guwahati Refinery

The Guwahati Refinery, located at Noonmati, Assam, is the first public sector oil refinery in India, established by Indian Oil Corporation Limited (IOCL). It was inaugurated on 1st January 1962 by the then Prime Minister of India, Pandit Jawaharlal Nehru, marking a historic milestone in India's energy sector.

Operated under the aegis of IOCL, Guwahati Refinery was built in collaboration with Rumanian assistance and started with an initial installed capacity of 0.75 million metric tonnes per annum (MMTPA). Over the decades, the refinery has undergone modernization and capacity enhancement, and as of 2025, it has a refining capacity of 1.2 MMTPA. Though small in size compared to other IOCL refineries, it plays a vital role in meeting the petroleum product requirements of the northeastern region of India. The refinery processes a blend of Assam crude oils such as Nahorkatiya, Geleki, and Moran. It produces a wide range of products including LPG, Motor Spirit (MS), High-Speed Diesel (HSD), Superior Kerosene Oil (SKO), Aviation Turbine Fuel (ATF), and various petrochemical feedstocks. Guwahati Refinery is equipped with modern units such as the Delayed Coker Unit (DCU), Diesel Hydro Treating (DHDT) unit, and other essential processing and utility facilities. It also houses departments such as Electrical, Mechanical, Instrumentation, P&U (Power & Utilities), and Fire & Safety, which ensure smooth and safe operations. The refinery follows strict environmental norms and has invested in effluent treatment and pollution control technologies to ensure compliance with sustainability goals. It has also adopted several green initiatives in alignment with IOCL's vision to reduce carbon footprint and move toward renewable energy. Despite being one of the oldest refineries, Guwahati Refinery has consistently demonstrated excellence in operational efficiency, safety, and innovation. Its strong focus on training, skill development, and employee welfare makes it a model unit within the IOCL group. For engineering students and interns, it offers valuable exposure to real-time industrial systems, advanced control technologies, safety protocols, and sustainable practices. The refinery continues to contribute significantly to the regional economy while supporting IOCL's mission of fueling India's energy needs.

IOCL GUWAHATI REFINERY, NOONMATI, ASSAM, INDIA

Substation at IOCL Guwahati Refinery

The electrical substation at IOCL Guwahati Refinery forms the backbone of the refinery's power distribution system, ensuring the reliable and uninterrupted supply of electricity required for continuous operation of all process units, utility systems, and safety infrastructure. Power plays a critical role in refinery operations, where high-capacity motors, heaters, pumps, control systems, and lighting loads must be operated with high reliability, safety, and precision. The substation is designed to meet these demands by stepping down high-voltage power from the grid or in-house captive generation to usable voltages and distributing it efficiently across the refinery through an integrated system of transformers, panels, and protection devices.

The incoming electrical supply to the Guwahati Refinery is generally fed at 132 kV, which is stepped down to 6.3 kV and further to 415 V / 230 V levels depending on the type of load and application. This transformation and distribution are carried out by power transformers and distribution transformers strategically located within the substation premises. The installed transformer capacities vary based on critical load requirements; typically, 132/6.3 kV power transformers are rated around 5 MVA to 10 MVA, while 6.3/0.415 kV distribution transformers can range from 250 kVA to 1 MVA. The substation also includes HT panels (High Tension), LT panels (Low Tension), VCB (Vacuum Circuit Breakers) for isolation and protection, Current Transformers (CTs), Potential Transformers (PTs), and protective relays to ensure safe and automatic shutdown in case of faults.

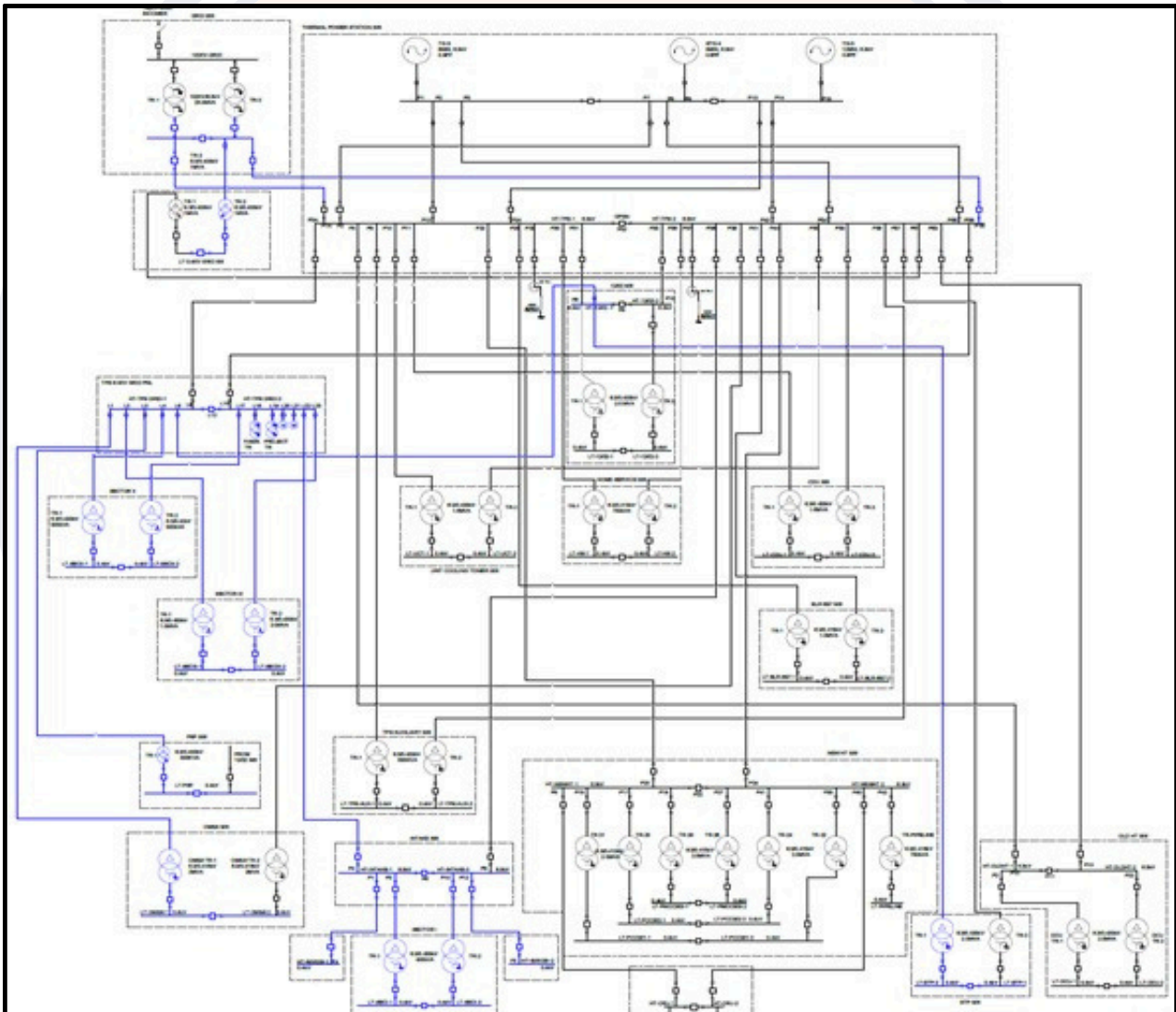
A major part of the substation's infrastructure revolves around control and distribution panels that manage and monitor power flow throughout the plant. These panels are:

1. PCC (Power Control Center): These panels handle the main power distribution of the plant at the LT level (415 V), supplying power to various MCCs, lighting panels, and utility loads. PCCs are usually equipped with Air Circuit Breakers (ACBs), energy meters, and protective relays.
2. MCC (Motor Control Center): MCCs control the operation of electric motors in the refinery, such as pumps, compressors, blowers, etc. These panels include DOL (Direct-On-Line), Star-Delta, and VFD (Variable Frequency Drive) starters along with protection devices like overload relays, fuses, and isolators.

3. PMCC (Power & Motor Control Center): A combination of PCC and MCC in a single unit, PMCC panels serve both motor and non-motor loads. These are often used in large areas with mixed load types and help optimize space and control logic.
4. ASB (Annunciation & Synchronization Board): Used for generator and load management. These panels display alarms and fault conditions and help in synchronizing multiple power sources (such as mains and DG sets).
5. DCDB (Direct Current Distribution Board): DCDB panels distribute DC power (usually 24V or 110V DC) for control circuits, relays, and instrumentation. These are essential for critical systems that require uninterrupted DC supply.
6. UPS (Uninterruptible Power Supply) Panels: UPS panels provide backup AC power to critical loads such as DCS/SCADA systems, emergency lighting, PLC panels, fire and gas systems, etc., in case of main power failure. The UPS system generally includes battery banks, inverters, and static switches.
7. LCP (Local Control Panels): Installed near field equipment, LCPs allow local manual control and feedback from motors or instruments. They serve as an interface between the control room and the field.
8. Relay & Protection Panels: These are used to protect the entire substation and downstream equipment from overcurrent, earth faults, under-voltage, and other faults. They typically house numeric relays, trip circuits, CT/VT input modules, and event loggers.
9. Lighting Distribution Panels: These panels supply power to industrial and office lighting across refinery zones and are typically isolated from critical process panels to avoid interference.

The entire electrical system is monitored from the Main Control Room (MCR) and Electrical Control Room, where SCADA (Supervisory Control and Data Acquisition) and DCS (Distributed Control System) platforms help in real-time monitoring, fault detection, load management, and automation. The substation also includes earthing systems, cable trenches, busbars, and fire protection systems in accordance with safety standards like IEC and IS codes. Fire safety in electrical installations is further supported by flameproof panels and regular IR thermography inspections.

HT SINGLE LINE DIAGRAM-NEW SUBSTATION- 132/6.3kV



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Key Components Shown

Incoming Source

- Two incomers from 132kV / 6.30kV transformers (T-101 and T-102)
- These transformers supply power to the main busbars

Busbar Arrangement

- A main bus configuration with a tie breaker
- Several feeder circuits are connected to this main bus

Major Equipment Symbols

- Circuit Breakers (CBs) – present at incoming and outgoing lines
- CTs (Current Transformers) and PTs (Potential Transformers)
- Isolators / Disconnectors
- Earthing Switches
- Relays and protection panels

Substations and Feeders Covered

1. Incoming Transformers (132/6.30kV)

- Transformer T-101 and T-102 feed the HT panel
- Each has CT/PT metering and circuit breaker protection

2. Outgoing Feeders

The SLD includes feeders supplying power to:

- TPS HT generating section
- Old HT substation
- DM plant / intake area
- Control rooms
- Process units such as CDU, VDU, MSQ, DHDS, etc.

3. Interconnection Panels

- A bus-coupler breaker ensures redundancy
 - There is a tie connection between two bus sections
-

4. Auxiliary Transformers (6.3/0.415kV)

- Used to feed LT (low-tension) panels and internal utilities
-

Protection and Safety Elements

- Protective relays such as OCR (Overcurrent Relay), EFR (Earth Fault Relay)
 - Breaker interlocks and control wiring shown with dotted lines
 - Metering circuits for current, voltage, and power
 - Earthing switches for maintenance safety
-

Notes Section (Top-Right of Drawing)

1. Transformer vector groups and earthing details
 2. Protection and metering is based on a 3-phase, 3-wire system
 3. CT and PT ratio information to be finalized during detailed engineering
 4. Designed according to IEC standards
-

Drawing Information (Bottom-Right)

- Title: HT Single Line Diagram – New Substation
- Client: Indian Oil Corporation Ltd., Guwahati Refinery
- Prepared by: Jacobs Engineering
- Drawing No.: 46LK-5100-00-EL-DWG-0002
- Date: 12.01.2005
- Revision: Rev-0
- Scale: Not to Scale

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Purpose of This SLD

The SLD is used to:

- Understand how electrical power is distributed across the refinery
- Assist in planning shutdowns, maintenance, and safety procedures
- Analyze fault conditions and coordinate protection systems
- Design and integrate SCADA or automation systems
- Ensure redundancy and reliability of power supply

Summary of the Diagram

This is a high-tension (HT) power distribution Single Line Diagram (SLD) for a new substation at Guwahati Refinery. It shows the overall layout of the electrical system and how power is distributed to different sections of the refinery.

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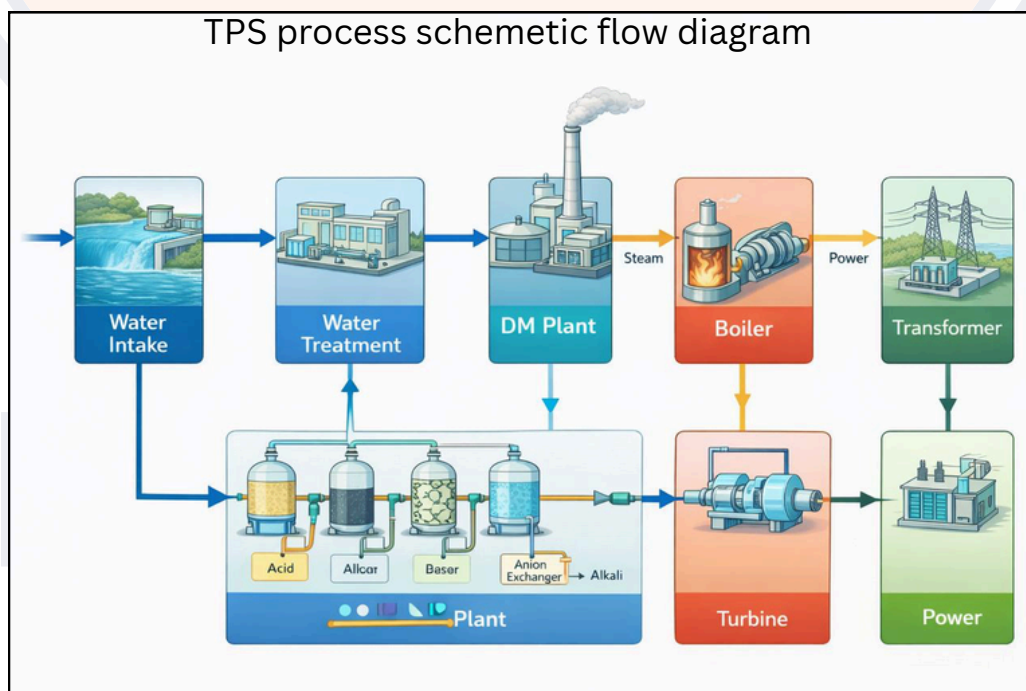
THERMAL POWER STATION (TPS)

A Thermal Power Station (TPS) is a power generation plant in which electrical energy is produced by converting thermal energy into mechanical energy and then into electrical energy. In a TPS, heat energy obtained from the combustion of fuel such as coal, oil, or gas is used to generate steam. This steam drives a turbine, which in turn rotates an alternator to produce electricity.

TPS form the backbone of electric power generation in many countries due to their reliability, ability to generate large amounts of power, and continuous operation. These plants are widely used to meet base load and peak load demands of industries and utilities.

TPS consist of major components such as intake system, water treatment plant, DM plant, boiler, steam turbine, generator, condenser, cooling system, and transformers. Efficient operation and maintenance of these systems ensure reliable power generation with high efficiency and safety.

Due to their capability to operate continuously and support large electrical loads, thermal power stations are extensively used in refineries, power plants, and large industrial complexes.



WATER TREATMENT PLANT (WTP)

1. A Water Treatment Plant (WTP) is a system designed to treat raw water and make it suitable for industrial use. In thermal power stations and refineries, WTP is an essential unit as it supplies treated water to various plant processes and acts as the first stage of water purification before further treatment in the DM plant. Raw water obtained from rivers, reservoirs, or bore wells contains suspended solids, turbidity, organic matter, and microorganisms. These impurities must be removed to protect downstream equipment and ensure smooth plant operation.

2. Purpose of WTP

The main purposes of a Water Treatment Plant are:

- Removal of suspended and colloidal impurities
- Reduction of turbidity and color
- Protection of DM plant and boiler systems
- Supply of clarified water for industrial processes

3. Major Processes in WTP

3.1 Coagulation and Flocculation

Chemicals such as alum are added to raw water to coagulate fine suspended particles and form flocs.

3.2 Sedimentation / Clarification

The flocs settle at the bottom of the clarifier, separating solid impurities from water.

3.3 Filtration

Water is passed through sand or multimedia filters to remove remaining fine particles.

3.4 Disinfection

Chlorination or other disinfection methods are used to kill harmful microorganisms.

4. Importance of WTP in TPS

- Ensures reliable supply of clean water
- Prevents fouling and scaling in equipment
- Improves efficiency of DM plant
- Reduces maintenance and operating problems

DM (DEMINERALIZATION) PLANT

A Demineralization (DM) Plant is a water treatment system used to remove dissolved salts, minerals, and ionic impurities from raw water. The water produced by a DM plant is known as demineralized water, which has very low electrical conductivity and is free from hardness-causing ions.

In refineries and power plants, DM water is essential for boilers, turbines, cooling systems, and laboratory applications, as impurities in water can cause corrosion, scaling, and damage to equipment.

2. Need for DM Water in Refineries

DM water is required in refineries for the following reasons:

- To prevent scaling in boilers and heat exchangers
- To avoid corrosion of pipelines and equipment
- To ensure safe and efficient boiler operation
- To improve equipment life and reliability
- To meet process and quality requirements

3. Raw Water Characteristics

Raw water used in DM plants may contain:

- Dissolved salts (chlorides, sulphates)
- Hardness (calcium, magnesium)
- Silica
- Dissolved gases (CO_2 , O_2)
- Suspended solids

These impurities must be removed before water is used in critical applications.

4. Principle of DM Plant Operation

The DM plant works on the principle of ion exchange. In this process, undesirable ions present in raw water are exchanged with hydrogen (H^+) and hydroxyl (OH^-) ions using ion exchange resins.

When H^+ and OH^- ions combine, they form pure water (H_2O).

5. Main Components of a DM Plant

5.1 Pretreatment Unit

Pretreatment improves raw water quality and protects downstream equipment.

Includes:

- Pressure sand filter – removes suspended solids
- Activated carbon filter – removes chlorine, organic matter
- Dosing systems – pH correction and chemical dosing

5.2 Cation Exchange Unit

In the cation exchanger, all positive ions such as calcium, magnesium, sodium, and iron are replaced by hydrogen ions.

Reaction example:



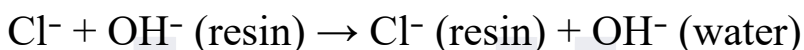
5.3 Degasser Unit

Degasser removes dissolved gases, mainly carbon dioxide (CO₂), which reduces load on the anion exchanger.

5.4 Anion Exchange Unit

In the anion exchanger, all negative ions such as chloride, sulphate, nitrate, and silica are replaced by hydroxyl ions.

Reaction example:



5.5 Mixed Bed Unit (Polisher)

The mixed bed contains a mixture of cation and anion resins and provides final polishing of DM water.

- Produces very low conductivity water
- Ensures high purity

5.6 Regeneration System

Ion exchange resins lose effectiveness over time and require regeneration.

- Cation resin is regenerated using acid (HCl or H_2SO_4)
- Anion resin is regenerated using alkali (NaOH)

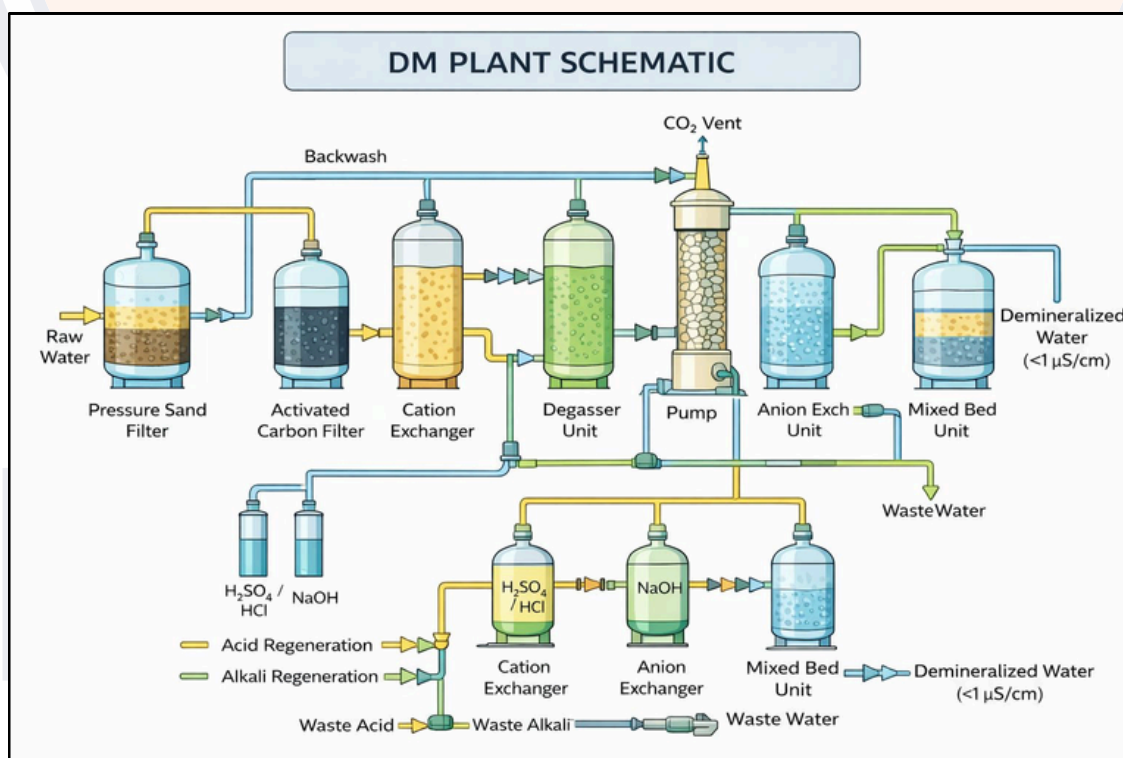
6. Quality of DM Water

Typical DM water characteristics:

- Conductivity: $< 1 \mu\text{S/cm}$
- Hardness: Nil
- Silica: Very low
- pH: Neutral to slightly acidic

7. Applications of DM Water in Refineries

- Boiler feed water
- Turbine systems
- Cooling water makeup
- Chemical preparation
- Laboratory use



BOILER

A boiler is a vital component of a thermal power station in which water is converted into high-pressure and high-temperature steam by the application of heat. The steam produced in the boiler is used to drive a steam turbine for power generation. Boilers are designed to operate safely and efficiently under high pressure and temperature conditions.

In thermal power stations, demineralized (DM) water is used as boiler feed water to prevent scaling, corrosion, and damage to boiler tubes and associated equipment.

2. Purpose of Boiler

The main purposes of a boiler are:

- To convert water into steam
- To generate steam at required pressure and temperature
- To supply continuous steam to the turbine
- To ensure efficient heat utilization

3. Basic Working of Boiler

Fuel is burned in the furnace, producing heat. This heat is transferred to water flowing through boiler tubes, converting it into steam. The steam is collected in the steam drum and then supplied to the turbine.

4. Importance of Boiler in TPS

- Core component for steam generation
- Determines efficiency of the power plant
- Supplies energy to drive the turbine
- Requires strict water quality control

THERMAL POWER STATION AT IOCL

GUWAHATI REFINERY

INTRODUCTION

The Thermal Power Station (TPS) at IOCL Guwahati Refinery was originally commissioned in 1962 with technological assistance from the Romanian Government. The initial setup included four Romanian boilers of 20 TPH (tons per hour) capacity each, two Romanian turbo-generators of 3 MW each, and a water softening plant to treat feed water for boilers and process use.

Over the past several decades, the TPS has undergone continuous modernization and expansion to meet the growing energy demands of the refinery and to support new high-load process units. The TPS now serves a dual purpose: electricity generation and process steam supply for refinery operations. The system uses multiple fuels such as fuel oil, refinery fuel gas, MRN (mixed refinery naphtha), and natural gas.

Steam is generated in high-capacity boilers, pressurized to approximately 37.5 kg/cm², and passed through a common steam header to drive steam turbines.

These turbines are mechanically coupled to alternators (generators) to produce electrical energy. Additionally, steam is extracted at controlled medium-pressure (MP) and uncontrolled low-pressure (LP) stages for process heating in refinery units.

Modernization Timeline and Key Developments

1985: Generator #3 commissioned (8 MW).

1986: Cooling system upgraded to recirculating type from once-through system.

1992: Boiler #5 (IJT make, 40 TPH) commissioned.

1993: Replacement of old water softening plant with modern DM (Demineralization) plant.

1998: Generator #4 commissioned.

2002: 3rd chain of DM plant installed.

2004: TG #3 and TG #4 upgraded from ECIL to ABB MMI systems.

2005: Generator #5 (12 MW) commissioned; TPS cooling tower renovated.

2006: STG #5 commissioned; Deaerator 1 & 2 removed; Romanian Boiler #2 decommissioned.

- 2007: Romanian Boiler #1 phased out.

As of 2025, the Thermal Power Station continues to operate with a mix of old and new equipment, focusing on energy efficiency, automation, and compliance with safety and environmental standards.

Major Equipment's in TPS – Guwahati Refinery (As of 2025)

Boilers:

- 2 × 20 TPH Romanian Boilers (Boilers #3, #4 – legacy units, partially operational/standby)
- 1 × 40 TPH IJT Boiler (Boiler #5 – main process steam contributor)
- 2 × 50 TPH Thermax Boilers (Boilers #6 and #7 – latest high-efficiency units)

Total Installed Steam Generation Capacity: 180 TPH

Steam Turbines (TG sets):

- 2 × 8 MW BHEL Make Extraction-cum-Condensing Turbines
- 1 × 12 MW BHEL Make Extraction-cum-Condensing Turbine

Total Installed Power Generation Capacity: 28 MW

DM (De-mineralization) Water Plant:

- 2 × 50 m³/hr Chains
- 1 × 60 m³/hr Chain

Total Treated Water Generation Capacity: 160 m³/hr

Cooling Towers:

- TPS Cooling Tower: Dedicated to power plant condenser systems
- Unit Cooling Tower: Serves CDU (Crude Distillation Unit) and DCU (Delayed Coking Unit)
- Process Cooling Tower: Supplies HGU (Hydrogen Generation Unit), HDT (Hydrotreater), INDMAX, and Nitrogen Unit

MAJOR ELECTRICAL EQUIPMENT DETAILS

Turbogenerator #3

Parameter	Specification
MAKE	BHEL
TYPE	1000
Continuous rating at Generator Terminals	MW
Turbine Speed	8000 RPM
Generator Speed	3000 RPM
Coupling	Speed Reduction Gear
Live Steam Pressure	35 ATA
Live Steam Temperature	435 °C
Max. Live Steam Flow	63 MT/hr
Speed Governor	Hydro-dynamic
Governing System	RIV
Pressure Governor	ASKANIA
Protection	Over speed, Axial Displacement, Low lube oil pressure, Low vacuum, High Vibration

Turbogenerator #4

Parameter	Specification
MAKE	BHEL
TYPE	1000
Continuous rating at Generator Terminals	8 MW
Turbine Speed	8000 RPM
Generator Speed	3000 RPM
Coupling	Speed Reduction Gear
Live Steam Pressure	35 ATA
Live Steam Temperature	435 °C
Max. Live Steam Flow	63 MT/hr
Speed Governor	Hydro-dynamic
Governing System	RIV
Pressure Governor	ASKANIA
Protection	Over speed, Axial Displacement, Low lube oil pressure, Low vacuum, High Vibration

MAJOR ELECTRICAL EQUIPMENT DETAILS

Turbogenerator #5

Parameter	Specification
MAKE	BHEL
TYPE	EK-100-2
Continuous rating at Generator Terminals	12 MW
Turbine Speed	6500 RPM
Generator Speed	1500 RPM
Coupling	Speed Reduction Gear
Live Steam Pressure	35 ATA
Live Steam Temperature	435 °C
Max. Live Steam Flow	94.7 MT/hr
Speed Governor	Electronic
Governing System	Electronic
Pressure Governor	ASKANIA
Protection	Over speed, Axial Displacement, Low lube oil pressure, Low vacuum, High Vibration

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Load Shedding

In large industrial power systems such as refineries and thermal power stations, load shedding is a critical protection and control function used to maintain system stability during abnormal conditions. Load shedding prevents complete system collapse by disconnecting selected loads when generation is insufficient or system parameters such as frequency and voltage fall beyond permissible limits.

A well-designed load shedding scheme ensures safe operation of generators, protects critical equipment, and enables quick plant revival after disturbances.

2. Need for Load Shedding

Load shedding is required under the following conditions:

- Sudden loss of generation (TG / STG trip)
- Grid failure or grid power becoming zero
- Sudden increase in load demand
- Severe frequency drop
- Islanding operation of plant

Without load shedding, these events can cause generator stalling, blackout, and equipment damage.

3. Load Shedding Philosophy (MOST IMPORTANT)

In the plant system, load shedding is implemented based on three main philosophies:

3.1 Frequency-Based Load Shedding (Under-Frequency Load Shedding – UFLS)

Frequency directly reflects the balance between generation and load. When load exceeds generation, frequency drops.

Operation

Load shedding stages are configured at different frequency levels with time delays.

Typical Stages (as per notes logic):

- Stage-1:
48.8 Hz for 0.1 sec
- Stage-2:
48.4 Hz for 0.5 sec
- Stage-3:
48.0 Hz for 2 sec
- Stage-4:
47.8 Hz for 3 sec

At each stage, pre-defined loads are tripped based on priority.

Purpose

- Arrest frequency decay
- Prevent generator trip
- Maintain system stability

3.2 Rate of Change of Frequency (df/dt) Based Load Shedding

This philosophy detects how fast the frequency is falling, not just the absolute value.

A rapid fall in frequency indicates severe generation-load imbalance.

Operation

- df/dt relay detects sudden frequency drop
- Immediate load shedding without waiting for frequency thresholds

Advantage

- Very fast response
- Prevents deep frequency collapse
- Acts as backup to UFLS

3.3 Priority-Based Load Shedding

All plant loads are categorized based on process criticality.

Priority Classification (from notes):

- Priority-1 → Essential / critical loads
- Priority-2 → Important process loads
- Priority-3 → Semi-critical loads
- Priority-4 → Non-essential loads (shed first)

Operation

During load shedding:

- Lowest priority loads are shed first
- Higher priority loads are retained as long as possible

Example (as per notes logic):

Total generation available = 16 MW

Available load after event = 10 MW

Load to be shed =

$$16 - 10 = 6 \text{ MW}$$

Loads are shed up to priority level required to remove 6 MW.

4. Load Shedding Scenarios Explained

Case 1: De-synchronization from Grid

When the plant gets de-synchronized from the grid, the system goes into island mode.

- Available generation reduces
- Immediate load shedding is required
- Load shedding is done as per priority to match generation

Case 2: Turbine Generator (TG) Trip

When a TG trips:

- Sudden loss of generation occurs
- Frequency starts dropping rapidly
- Frequency-based and df/dt load shedding operates
- Loads are shed to stabilize remaining generators

Overview of the Electrical Power System

The plant electrical system operates with a combination of captive generation (STGs/TGs) and grid supply. Power is distributed through 6.6 kV / 11 kV / 415 V bus systems, feeding various process and utility loads.

Under normal conditions, the plant may operate in:

- Synchronized mode with grid, or
- Island mode (grid unavailable)

The system is designed to maintain continuity of supply, protect generators, and avoid total blackout through automatic load shedding schemes.

2. Normal Synchronized Operation (Grid + Captive Generation)

In synchronized operation:

- Steam Turbine Generators (STGs) generate power (example: STG-4 = 12 MW)
- Grid import/export (example: Grid = 4 MW)
- Total system load = 16 MW

This is represented as:

STG + Grid → Synchronization → Load Supply

The synchronization conditions checked before closing breaker are:

- Voltage magnitude same
- Phase sequence same
- Phase angle difference $\approx 0^\circ$
- Frequency nearly equal (generator slightly higher)

Once synchronized, the system operates smoothly with automatic load sharing.

3. De-Synchronization (De-Synch) Operation

De-synchronization occurs when:

- Grid voltage dips
- Grid frequency fluctuates
- Grid fault occurs
- Intentional islanding is required

In normal de-synch operation, the plant continues to operate using captive generation only, but total available power reduces.

Example

- Before de-synch: 12 MW (STG) + 4 MW (Grid) = 16 MW
- After de-synch: Only 12 MW STG available

Hence, 4 MW load must be shed automatically.

POWER SYSTEM PROTECTION – BASIC COMPONENTS

1. Voltage transformers and current transformers:

To monitor and give accurate feedback about the healthiness of a system.

2. Relays:

To convert the signals from the monitoring devices, and give instructions to open a circuit under faulty conditions or to give alarms when the equipment being protected is approaching towards possible destruction.

3. Fuses:

Self-destructing to save the downstream equipment being protected.

4. Circuit breakers:

These are used to make circuits carrying enormous currents, and also to break the circuit carrying the fault currents for a few cycles based on feedback from the relays.

5. DC batteries:

These give uninterrupted power source to the relays and breakers that is independent of the main power source being protected.

DOL STARTER

A Direct-On-Line (DOL) starter is the simplest and most widely used motor starting method in industrial installations, including oil refineries. It is mainly employed for low and medium power induction motors where high starting current does not adversely affect the power system or mechanical load.

A DOL starter consists of both power circuit components and control circuit components, housed inside a metal enclosure suitable for industrial environments.

1. Power Circuit Components

1. Incoming Supply Terminals

Three-phase supply (R, Y, B) is fed into the starter from the MCC or PMCC panel.

2. Main Isolator / MCB / MCCB

Provides manual isolation and short-circuit protection for the motor circuit.

In refinery installations, MCCBs are preferred for higher fault-level withstand capability.

3. Contactor

- It consists of main power contacts and an electromagnetic coil.
- When the coil is energized, the main contacts close and full line voltage is applied directly to the motor.

4. Thermal Overload Relay (OLR)

Connected in series with the motor to protect against overload conditions.

- It operates on the heating effect of current.
- In case of prolonged overload, it trips and opens the control circuit, de-energizing the contactor.

5. Motor Terminals

The output of the starter is connected to the motor terminals (U, V, W)

2. Control Circuit Components

1. Control Supply

Usually derived from the same supply or through a control transformer (e.g., 415 V to 110 V AC).

2. Start Push Button (NO)

When pressed, it energizes the contactor coil, closing the main contacts.

3. Stop Push Button (NC)

Used to de-energize the contactor coil and stop the motor safely.

4. Auxiliary Contacts of Contactor

Used for self-holding (latching) of the contactor so that the motor continues running after releasing the start button.

5. Overload Relay Trip Contact (NC)

Opens the control circuit when overload occurs, stopping the motor.

6. Indication Lamps

Used to indicate motor ON, OFF, or TRIP status for operator awareness.

TRANSFORMER PROTECTION

A transformer is one of the most critical and costly components in a power system. It operates continuously and is subjected to electrical, mechanical, and thermal stresses. Any fault in a transformer can lead to serious damage, long outages, fire hazards, and heavy financial losses. Therefore, effective transformer protection is essential to ensure system reliability, safety of personnel, and continuity of power supply.

Transformer protection aims to detect internal and external faults quickly and isolate the faulty transformer from the system before severe damage occurs.

2. Need for Transformer Protection:

- Transformers are expensive and difficult to replace
- Internal faults can cause fire or explosion
- Faults may spread to connected equipment
- Long repair time leads to production loss
- Safety of operating personnel must be ensured

3. TYPES OF TRANSFORMER FAULTS

Transformer faults are broadly classified into internal faults and external faults.

a) Internal Faults

- Winding short circuits (phase-to-phase or phase-to-earth)
- Inter-turn winding faults
- Core faults due to insulation failure
- Overheating due to cooling system failure
- Tap changer faults

b) External Faults

- Short circuits on transmission lines
- Overloading conditions
- Through faults
- Over-voltage due to lightning or switching surges

4. TRANSFORMER PROTECTION SCHEMES

1. Differential Protection

Differential protection is the main protection for transformers.

It works on the principle of comparing the current entering and leaving the transformer. Under normal conditions, the difference is zero. During internal faults, the difference exceeds a set value and the relay operates.

Advantages:

- Fast operation
- Highly selective
- Sensitive to internal faults

2. Buchholz Relay Protection

Buchholz relay is used in oil-immersed transformers with a conservator tank.

Principle:

The relay is installed in the pipe connecting the main tank and the conservator. When an internal fault occurs, the transformer oil decomposes due to heat and produces gas. This gas accumulates in the Buchholz relay chamber.

Minor faults cause slow gas accumulation, while severe faults cause a sudden oil surge towards the conservator.

Operation:

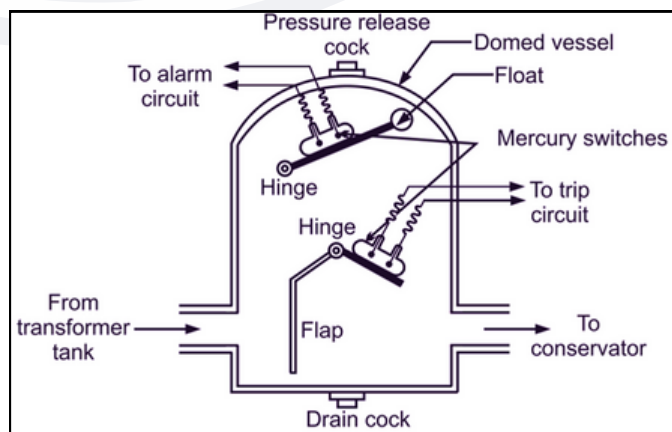
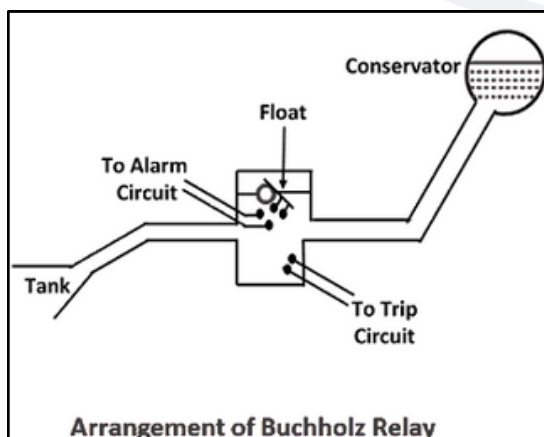
- Alarm contact: Operates during minor internal faults
- Trip contact: Operates during severe faults like winding short circuits

Faults Detected:

- Incipient faults
- Core heating
- Insulation failure
- Oil leakage

Advantages:

- Early fault detection
- Simple and reliable
- Prevents major transformer damage



3. Overcurrent Protection

Overcurrent protection is mainly used as backup protection for transformers.

Principle:

This scheme operates when the current flowing through the transformer exceeds a preset value for a specified time. Overcurrent relays are generally installed on the secondary side.

Purpose:

- Protection against external short circuits
- Protection against excessive overloads

Limitations:

- Cannot differentiate between internal and external faults
- Slower compared to differential protection

4. Earth Fault Protection

Earth fault protection is provided to detect insulation failure between transformer winding and earth.

Types:

(a) Restricted Earth Fault (REF) Protection:

This scheme provides sensitive protection for earth faults occurring within a specific zone of the transformer winding. It operates faster and is more sensitive than conventional earth fault protection.

(b) Unrestricted Earth Fault Protection:

Used to detect earth faults anywhere in the system, including external zones.

Advantages:

- High sensitivity
- Fast operation for earth faults
- Minimizes insulation damage

5. Overvoltage Protection

Transformers are vulnerable to overvoltages caused by lightning strikes and switching operations.

Protection Method:

- Lightning arresters are installed near transformer terminals
- Surge protection devices limit the magnitude of overvoltage

Importance:

- Prevents insulation breakdown
- Enhances transformer life

6. Temperature Protection

Excessive temperature severely affects transformer insulation life.

Protection Scheme:

- Oil temperature indicators monitor oil temperature
- Winding temperature indicators monitor conductor temperature

When temperature exceeds safe limits, alarms are issued or the transformer is tripped to prevent damage.

7. Pressure Relief Protection

During severe internal faults, pressure inside the transformer tank increases rapidly.

Function:

- Pressure relief device releases excess pressure
- Prevents tank rupture and explosion

GENERATOR PROTECTION

1. Introduction:

Generators are critical components of a power system, responsible for converting mechanical energy into electrical energy. In industrial plants, refineries, and power stations, generators operate continuously under varying electrical and mechanical stresses. Any fault or abnormal operating condition in a generator can lead to severe damage, long outages, safety hazards, and substantial financial losses.

Generator protection refers to the coordinated use of protective schemes designed to detect faults and abnormal conditions and disconnect the generator from the system before damage occurs.

2. Need for Generator Protection:

Generator protection is essential due to the following reasons:

- High cost and long replacement time of generators
- Possibility of severe mechanical and electrical damage
- Risk of fire and explosion
- Protection of turbine-generator shaft system
- Ensuring stability and reliability of the power system

Generator Protection Schemes

Generator protection employs multiple schemes to ensure complete coverage, as no single protection can detect all faults.

1. Stator Differential Protection

Stator differential protection operates on the principle of current comparison. Current transformers are installed at both ends of the stator winding. Under normal operating conditions, the currents entering and leaving the stator are equal.

During an internal stator fault, the differential current exceeds the relay setting, causing immediate tripping of the generator circuit breaker.

Faults Detected:

- Phase-to-phase faults
- Phase-to-earth faults
- Internal winding faults

Importance:

- Main protection for stator windings
- Fast and highly selective
- Prevents catastrophic damage

2. Earth Fault Protection

(a) Stator Earth Fault Protection:

Detects insulation failure between stator winding and earth using neutral grounding or residual current measurement.

(b) Rotor Earth Fault Protection:

Detects earth faults in the rotor field winding. While a single rotor earth fault may not cause immediate damage, a second fault can result in severe rotor damage.

3. Overcurrent Protection

Overcurrent protection operates when generator current exceeds a preset value for a specified time.

Purpose:

- Backup protection for external faults
- Protection against abnormal loading

Limitation:

- Not sensitive to internal generator faults

4 .Reverse Power Protection

Reverse power protection prevents the generator from operating as a motor when prime mover input fails but excitation remains.

Effects Prevented:

- Mechanical damage to turbine
- Overheating of generator
- Fuel wastage

5.Loss of Excitation Protection

Loss of excitation causes the generator to draw reactive power from the system and operate as an induction generator.

Effects:

- Rotor overheating
- Voltage instability
- System disturbance

Loss of excitation protection detects abnormal impedance or reactive power flow and isolates the generator.

6 .Negative Phase Sequence Protection

Unbalanced loads produce negative sequence currents, inducing double-frequency currents in the rotor.

Effects:

- Excessive rotor heating
- Mechanical stress
- Reduced machine life

7.Under-Frequency Protection

Under-frequency protection operates when system frequency falls below a safe limit due to overload or generation deficiency.

Purpose:

- Prevents turbine-generator instability
- Supports system stability
- Initiates load shedding if required

8.Overvoltage Protection

Overvoltage may occur due to sudden load rejection, excitation system failure, or switching surges.

Effects Prevented:

- Insulation stress
- Damage to connected equipment

9.Temperature Protection

Monitoring:

- Stator winding temperature
- Rotor temperature
- Bearing temperature

Excessive temperature results in alarm or tripping to prevent insulation deterioration.

10. Pole Slip Protection (Out-of-Step Protection)

Pole slip protection safeguards synchronous generators against loss of synchronism with the power system. During pole slip, the rotor no longer rotates in synchronism with the stator magnetic field, leading to severe mechanical and electrical stresses.

The protection operates by monitoring the impedance trajectory at the generator terminals. During loss of synchronism, the impedance moves in a characteristic path on the R-X plane. When this trajectory crosses predefined relay zones within a specified time, pole slip is detected.

Causes of Pole Slip:

- Severe system faults
- Sudden load changes
- Weak grid conditions
- Low or failed excitation
- Delayed fault clearing

Protection Action:

- Trips the generator circuit breaker
- Isolates the generator from the grid
- Prevents shaft, rotor, and coupling damage

Pole slip protection is an integral part of generator protection and is classified under protection against abnormal operating conditions.

Importance of Generator Protection in Industrial Plants

In refineries and power plants:

- Continuous power supply is critical
- Generator failure can shut down the entire plant
- Mechanical damage can be catastrophic
- Safety hazards are severe

Hence, reliable generator protection is mandatory.

RELAYS

A relay is a crucial protective device designed to detect abnormal conditions in an electrical circuit and initiate a trip signal to the associated circuit breaker, thereby isolating the faulty section to prevent damage. In earlier setups at Guwahati Refinery, various electromechanical relays were deployed for protection. However, with advancements in protection and control systems, these traditional relays are being gradually replaced with modern numerical relays.

Numerical relays offer improved reliability, faster operation, higher sensitivity, better fault analysis, and easier configuration. These relays incorporate microprocessors and digital signal processing techniques to monitor system conditions and execute precise protection algorithms. At Guwahati Refinery, several types of numerical relays have been installed to protect vital electrical assets, ensuring enhanced operational safety and system integrity.

Siemens make relay (installed in new units HT substation and 33 kV Switchgear)

- 7SJ600 Over Current Relay
- 7RW6000 Over/undervoltage relay
- 7VK512 Check synchronization relay
- 7UT512 Differential relay

ABB make relay (installed in new unit substation)

- SPJ30UC Over/undervoltage relay
- SPAD346 C2 differential relay

Alstom make relay (installed at 6.3 kV generation bus alstom section)

- Micom P-121, 127, 221, 921, 111, 211

CS PC

- PL300NC (CDV relay)

Easun Reyrolle

- RHO-3 motor protection relay
- Under frequency relay

CIRCUIT BREAKER

A Circuit Breaker (CB) is an essential switchgear device used to make or break an electrical circuit either manually during normal operation or automatically under fault conditions. It functions by separating its contacts—one fixed and one moving—when a fault occurs. Upon detection of a fault, the trip coil of the circuit breaker is energized, causing the moving contact to separate from the fixed contact, thereby interrupting the current flow. This interruption process generates an arc between the contacts, which must be extinguished quickly and safely.

This arc interruption is achieved using different arc-quenching media depending on the type of circuit breaker. Oil circuit breakers, air circuit breakers, and vacuum circuit breakers are commonly used in substations. In Guwahati Refinery, various types of circuit breakers are employed across the substation and field installations to ensure rapid fault isolation, system stability, and personnel safety.

1) Air Circuit Breaker (ACB):

An Air Circuit Breaker (ACB) operates by interrupting the circuit through contacts placed in open air. When a fault occurs and the contacts separate, an arc is formed which is divided into several smaller arcs by an arc chute. These smaller arcs are then quickly extinguished due to the cooling effect of air and electromagnetic forces.

ACBs are generally used in low-voltage applications and are commonly installed for systems operating in the voltage range of 380V to 480V. They provide effective protection against overcurrent, short-circuit, and ground faults in industrial installations like Guwahati Refinery.

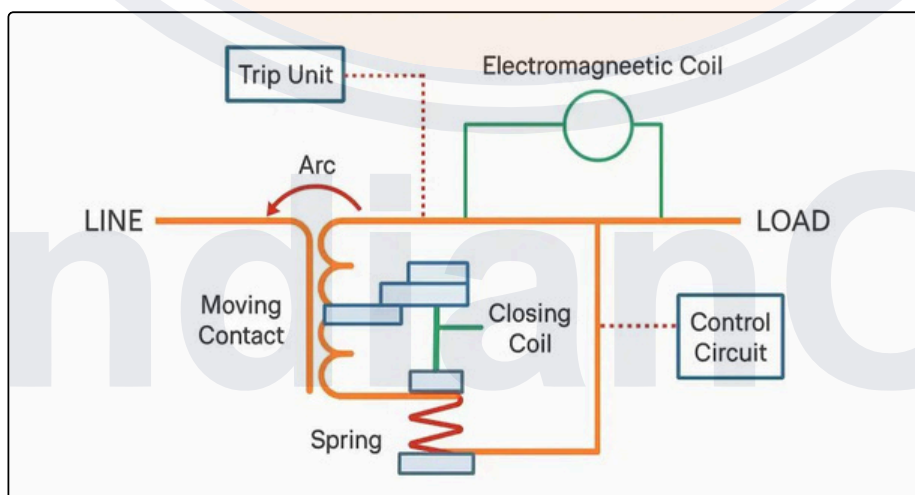


Fig. Air Circuit Breaker (ACB)

Advantages of Air Circuit Breaker (ACB)

- Uses air as the arc-quenching medium, so there is no risk of fire or explosion
- Suitable for high current ratings at low voltage
- Easy inspection and maintenance since contacts are accessible
- Provides adjustable protection settings (overload, short-circuit, earth fault)
- Good reliability for frequent switching operations
- Environment-friendly (no oil or SF₆ gas)
- Can be easily integrated with electronic/relay-based protection systems

Applications of Air Circuit Breaker (ACB)

- Used in Low Voltage (LV) systems up to 690 V
- Installed as incomer, feeder, bus-coupler, and tie breaker in switchboards
- Widely used in:
 - Power Control Centers (PCC)
 - Motor Control Centers (MCC)
 - Power plants and substations
 - Industrial plants and refineries
- Used for protection of:
 - Large motors
 - Generators
 - Transformers (LV side)
 - Busbars and distribution feeders

Vacuum Circuit Breaker (VCB)

A Vacuum Circuit Breaker (VCB) is a type of circuit breaker used mainly in medium-voltage systems (typically 3.3 kV to 33 kV). In a VCB, arc quenching takes place in a vacuum, which has very high dielectric strength.

When a fault occurs, the breaker contacts separate inside a vacuum interrupter. An arc is formed momentarily, but since there is no ionized medium in vacuum, the arc is extinguished very quickly, usually at the first current zero. This results in fast and reliable interruption of the fault current.

The vacuum inside the interrupter is maintained at about 10^{-6} to 10^{-8} bar, which ensures excellent insulation and rapid arc extinction. Due to this, VCBs have high breaking capacity, long service life, and minimal maintenance requirements.

Main Components of a VCB

- Fixed and moving contacts
- Vacuum interrupter
- Metallic bellows (to allow movement while maintaining vacuum)
- Insulating enclosure
- Operating mechanism

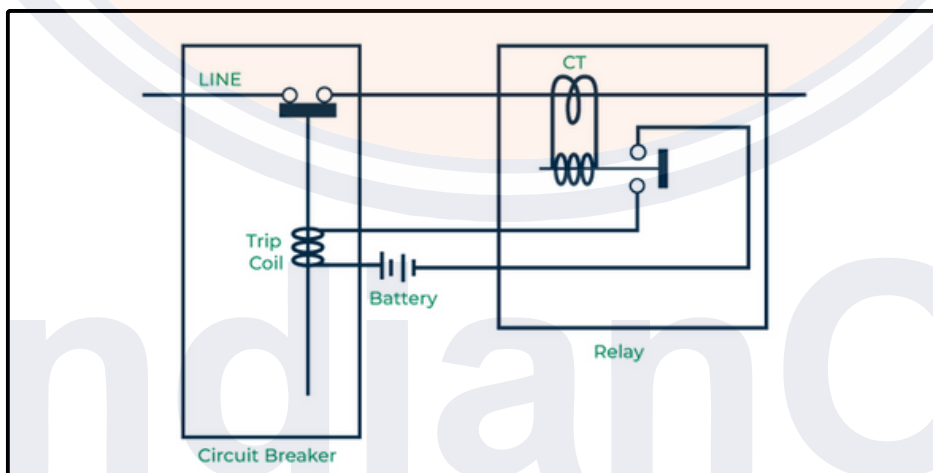


Fig: Vacuum Circuit Breaker (VCB)

Advantages of VCB

- Fast arc extinction
- Low maintenance
- High reliability
- Compact size
- Environment-friendly (no oil or gas)

Applications

VCBs are widely used in HT and MV substations, industrial plants, power stations, and utilities for protection against overcurrent, short circuits, and earth faults.

Applications of VCBs at Guwahati Refinery

In Guwahati Refinery, Vacuum Circuit Breakers (VCBs) are widely deployed in various high-tension (HT) substations due to their reliability and low maintenance requirements. The specific substations using vacuum circuit breakers include:

- All HT breakers of the new substation (Make: Siemens)
- All HT breakers in the Thermal Power Station (TPS) HT generating section (Make: Areva and Alstom)
- All HT breakers at the old HT substation (Make: Jyoti)
- All HT breakers in the new intake substation (Make: Areva)

These VCBs ensure safe and efficient operation of the electrical system under both normal and fault conditions.

GS CIRCUIT BREAKER

A 132 kV GS (circuit breaker) is a high-voltage switching and protection device used in transmission and sub-transmission systems. It makes, carries, and interrupts normal load current as well as fault current under both normal and abnormal system conditions.

Construction of 132 kV GS Circuit Breaker

The major components of a 132 kV GS circuit breaker are:

1. SF₆ Gas Chamber

The interrupting medium is Sulphur Hexafluoride (SF₆) gas, which has excellent dielectric strength and arc-quenching properties. The gas is filled at a specified pressure and sealed within the breaker enclosure.

2. Fixed and Moving Contacts

- The fixed contact remains stationary.
- The moving contact operates during opening and closing.

3. Arc Interruption Unit

Modern GS breakers use puffer or self-blast mechanisms where SF₆ gas is compressed or directed across the arc to extinguish it rapidly.

4. Operating Mechanism.

It provides the mechanical energy required for fast opening and closing of breaker contacts during fault conditions.

5. Insulators and Bushings

Porcelain or composite insulators provide mechanical support and electrical insulation between live parts and earth.

6. Gas Density Monitor and Alarm

Continuously monitors SF₆ gas pressure.

Alarms and lockouts are provided to prevent operation at unsafe gas pressure levels.

7. Control and Relay Panels

House trip coils, closing coils, auxiliary relays, interlocks, and protection interfaces.

Working Principle

Under normal conditions, the breaker contacts remain closed, allowing current to flow.

When a fault occurs, the protection relay sends a trip signal to the breaker trip coil. The operating mechanism separates the moving contact from the fixed contact, and an arc is formed.

SF₆ gas rapidly absorbs free electrons and cools the arc, increasing dielectric strength and interrupting the fault current at the first current zero. The arc is extinguished quickly, ensuring safe isolation of the faulty section.

Applications in Oil Refinery Power System

- Incoming grid feeders
- Power transformers (132/33 kV or 132/11 kV)
- Transmission line protection

They ensure high reliability, system stability, and safety for critical refinery operations.

UPS (UNINTERRUPTED POWER SUPPLY)

1. Introduction

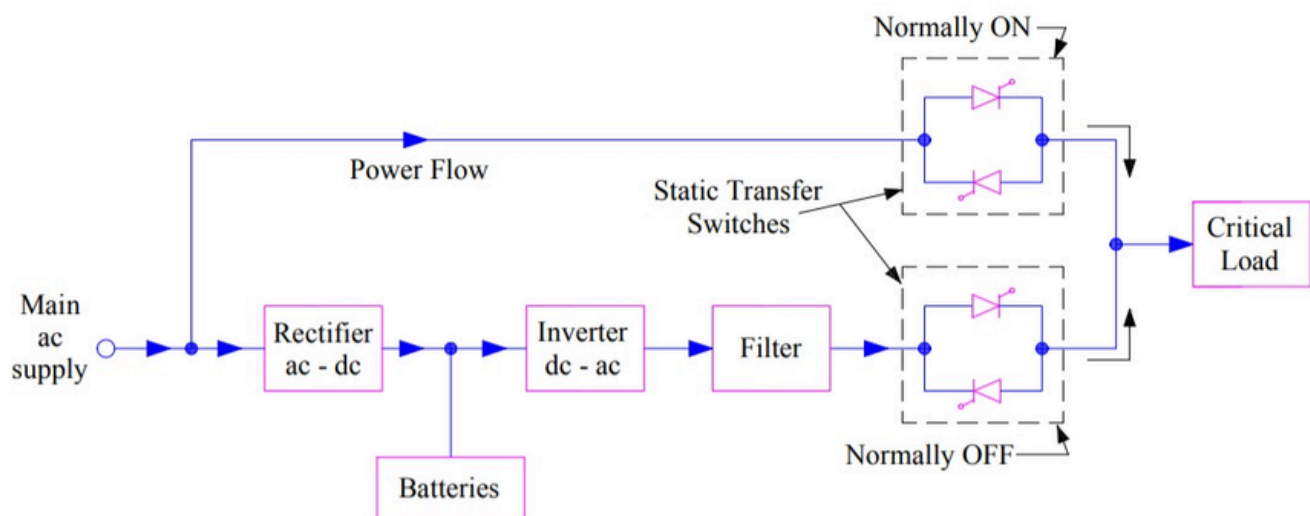
An Uninterruptible Power Supply (UPS) is an electrical system designed to provide continuous and reliable power to critical loads when the main power supply fails or becomes unstable. A UPS ensures uninterrupted operation of sensitive equipment by supplying power from stored energy sources such as batteries or flywheels.

UPS systems are widely used in refineries, power plants, hospitals, data centers, control rooms, PLC/DCS systems, and communication networks, where even a momentary power interruption can lead to serious operational and financial losses.

2. Need for UPS

UPS systems are required for the following reasons:

- To maintain continuity of power during outages
- To protect sensitive equipment from voltage fluctuations
- To prevent data loss and system crashes
- To ensure safe shutdown of critical systems
- To improve power quality and reliability



Arrangement of UPS

3. Working Principle of UPS

Under normal conditions, the UPS receives AC power from the mains. The rectifier converts this AC power into DC, which charges the battery and feeds the inverter. The inverter continuously supplies clean and stable AC power to the load.

When the mains supply fails or goes beyond acceptable limits, the UPS instantly supplies power from the battery through the inverter, ensuring zero or minimal interruption to the load.

When mains power is restored, the UPS switches back to normal operation and recharges the batteries.

4. Basic Components of a UPS

A UPS system consists of several major components that work together to ensure uninterrupted and high-quality power supply to critical loads.

1 Rectifier

The rectifier converts incoming AC power from the mains into DC power. This DC power serves two purposes:

- Supplying DC input to the inverter
- Charging the battery bank in a controlled manner

Modern rectifiers use thyristor-based or IGBT-based circuits with power factor correction to improve efficiency and reduce harmonics. The charger section regulates charging current and voltage to prevent battery overcharging or overheating.

2 Battery Bank

The battery bank is the heart of the UPS system, as it provides stored energy during power failures.

The battery bank stores electrical energy in chemical form and supplies DC power to the inverter when the mains supply fails or goes outside acceptable limits. This ensures uninterrupted power to critical loads.

Construction:

A battery bank consists of multiple battery cells connected in series and/or parallel to achieve the required DC voltage and capacity. Typical UPS battery voltages range from 110 V DC, 220 V DC, or higher, depending on system size.

Importance

- Ensures continuity of power during outages
- Determines reliability of the UPS
- Failure of the battery bank renders the UPS ineffective

3 Inverter

The inverter converts DC power from the battery bank or rectifier into stable, regulated AC power suitable for the load.

Modern inverters use PWM (Pulse Width Modulation) techniques to produce a sinusoidal output with low harmonic distortion. The inverter ensures constant voltage and frequency irrespective of input variations.

4 Static Switch

The static switch is a fast electronic switching device that transfers the load between the inverter and bypass supply.

Function

- Transfers load to bypass during inverter failure or overload
- Ensures seamless operation with no interruption

Static switches use thyristors or IGBTs and operate within milliseconds.

5 Bypass Circuit

The bypass circuit provides an alternate path for power to supply the load directly from the mains.

Types

- Manual bypass – used during maintenance
- Automatic bypass – operates during UPS fault or overload

Bypass ensures continuity of supply even when the UPS is unavailable.

ELECTRICAL SAFETY IN REFINERIES

Refineries handle large quantities of flammable gases, vapours, and liquids, making them highly hazardous industrial environments. Any electrical spark, overheating, or arc can lead to fire or explosion. Therefore, electrical safety is of utmost importance in refineries.

Electrical safety in refineries focuses on hazardous area classification, selection of explosion-protected equipment, and strict adherence to safety standards to prevent ignition of explosive atmospheres.

1. Hazardous Areas in Refineries

A hazardous area is a location where flammable gases, vapours, or combustible dust may be present in sufficient quantity to form an explosive atmosphere.

Common Hazardous Locations in Refineries:

- Process units
- Pump houses
- Compressor areas
- Storage tanks and tank farms
- Loading and unloading areas

2. Hazardous Area Classification (Zones)

Hazardous areas are classified based on the frequency and duration of presence of flammable gases.

Zone 0

- Explosive gas atmosphere present continuously or for long periods
- Example: Inside storage tanks

Zone 1

- Explosive gas atmosphere likely during normal operation
- Example: Pump seals, valve areas

Zone 2

- Explosive gas atmosphere unlikely in normal operation and present only for short duration
- Example: Surrounding areas of Zone 1

3. Gas Groups

Flammable gases are classified into gas groups based on their ignition characteristics. This classification helps in selecting suitable electrical equipment.

Gas Groups Used in Refineries:

- Group IIA – Less hazardous gases (e.g., propane)
- Group IIB – More hazardous gases (e.g., ethylene)
- Group IIC – Highly hazardous gases (e.g., hydrogen, acetylene)

Equipment approved for IIC is safe for IIB and IIA as well.

4. Explosion-Protected Electrical Equipment

To ensure safety, special types of electrical equipment are used in hazardous areas.

5. Flameproof Equipment (Ex d)

Flameproof equipment is designed so that any internal explosion is contained within the enclosure and does not ignite the surrounding atmosphere.

Flameproof Lights:

- Used extensively in refinery hazardous zones
- Strong, heavy enclosures
- Flame paths prevent flame propagation
- Marked as Ex d

Applications:

- Process areas
- Pump houses
- Tank farms

Maintenance of Electrical Equipment and Electrical Control Panels

1. Importance of Maintenance

Proper maintenance of electrical equipment and control panels is essential to ensure safe, reliable, and uninterrupted operation. It helps in preventing unexpected failures, reducing unplanned shutdowns, extending equipment life, improving energy efficiency, and complying with safety and statutory standards.

2. Types of Maintenance

a) Preventive Maintenance (PM)

Planned and periodic inspections carried out to minimize breakdowns.

Key activities include tightening electrical connections, cleaning panels and equipment, lubricating moving parts, thermal scanning for hot spots, and testing protective devices such as relays and circuit breakers.

b) Predictive Maintenance

Uses condition-monitoring techniques to predict failures in advance.

Common methods include vibration analysis, infrared thermography, oil analysis, and insulation resistance testing.

c) Corrective Maintenance

Performed after a failure occurs and involves fault diagnosis, repair, or replacement of defective components.

3. Maintenance of Electrical Equipment

Transformers

- Check oil level and dielectric strength
- Inspect bushings for cracks or contamination
- Measure winding resistance and insulation resistance
- Monitor load and temperature conditions

Motors (HT / LT / Induction)

- Clean terminals and motor body
- Check bearing lubrication and alignment
- Measure insulation resistance
- Observe vibration, noise, and overheating

Circuit Breakers (ACB / VCB / OCB)

- Clean and inspect contacts
- Check arc chutes and contact wear
- Test mechanical operation and trip mechanism
- Lubricate moving parts

Relays

- Check pickup and drop-off values
- Perform secondary injection testing
- Verify settings with protection coordination

Switchgear

- Inspect for dust, corrosion, and loose parts
- Check earthing and interlocks
- Test busbar insulation resistance

4. Maintenance of Electrical Control Panels

Control panels contain protection, control, metering, and signaling equipment.

Key Maintenance Steps

- Ensure proper power isolation before work
- Clean panels using dry methods only
- Inspect wiring and tighten terminals
- Perform insulation (megger) testing
- Verify operation of relays, MCBs, MCCBs, and contactors
- Check indicators, selector switches, push buttons, alarms, and interlocks
- Test manual and auto operating modes.

CONCLUSION

The one-month internship at IOCL Guwahati Refinery, held from 15 Dec 2025 to 15 Jan 2026, provided a highly enriching and practical learning experience. Guwahati Refinery, often referred to as the “Gangotri of Indian Oil”, stands as a symbol of India’s industrial excellence and has contributed immensely to the nation’s refining sector since its inception.

This internship offered in-depth exposure to the Electrical Department and Power & Utilities (P&U) sections, covering various aspects such as substation operations, transformer and motor protection, thermal power station processes, and overall electrical safety systems. The refinery has already incorporated several modern technologies, including numerical relays, advanced circuit breakers, and intelligent motor protection relays, enhancing operational efficiency and system reliability.

However, modernization efforts—particularly in the Thermal Power Station (TPS)—are crucial to further improve system efficiency and replace outdated infrastructure. The implementation of smart switchgear, IoT-based monitoring, automated solutions, and predictive maintenance practices can significantly help the refinery align with global energy efficiency standards.

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This internship has been a valuable milestone in my professional development, and I am truly thankful to IOCL Guwahati Refinery for offering me this opportunity to learn and grow in a real-world industrial environment.